

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Understanding the Basics of Agile Software Development



Learning Outcome:

Describe ways of solving problems and getting things done in written and spoken situations and in work and social contexts.

Train the system users on application structuring and management, according to the established plan and user profile, according to the organization policies.

Can Do Statements:

- I can say what Agile software development is.
- I can list an overview of characteristics for Agile software development.
- I can explain specific information on Agile software development and its relation to software management.

Ask your partner:

- Do you think that the methodology used for developing software affects the performance of that development process? Why?
- What methodologies for developing software do you know?
- What processes or steps are the most important when developing software?





Let's Explore!

Let's dig into* Agile Software Development!

* We use the expression 'dig into' when we want to look into something or investigate or learn about a subject.



Instructions: Read the text below.

- Mary:** I read the manual about agile software development, and I wanted to review some concepts with you. So, this method describes the principles for software development in which flexibility for easily adapting to changes and collaboration between the team members are at its core, doesn't it?
- Luis:** In general terms, yes. I understood that the agile methodology is iterative, which means that the project is divided into short time frames. Then, by the end of each iteration, the products are tested to know where improvement is needed, aren't they?
- Mary:** I think you are right. Also, I read that as the demands of the market change constantly, and they require products to be ready promptly, companies have had to find new methods for software management. Consequently, effectiveness and flexibility are central in these new methods. Now, for this methodology, there aren't many requirements at the beginning of the project, are there?
- Luis:** Sometimes the customers present requirements and specifications of what they want, but they may request some changes afterward, and the team could resolve to apply other modifications thanks to feedback loops. This is what's called incremental development. There are some cases in which the client has only an idea of what he wants and as the project advances, the product and the final goal evolve along. I think Mr. Hardy called this as an evolutionary feature, didn't he?
- Mary:** Exactly! He also talked about the quality focus of this methodology. He said that agile method allows the resulting product to be more satisfying since it went through various testing phases and planning adaptations. I think software managers should consider using this method as it encourages the interaction of the team, the true requirements of the customer and the confidence in the quality of the product, shouldn't they?



Let's Practice!

Mary summarizes some characteristics for Agile Software development.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: more satisfying after / encourages the interaction / is iterative / has had a quality / flexibility for easily.

1. On this method, _____ adapting to collaboration between team members is at its core.
2. The Agile Methodology _____, so the project is divided into short time frames.
3. This methodology _____ focus.
4. Agile method allows the resulting product to be _____ various testing phases and adaptations.
5. It _____ of the team and the confidence in the quality of the product.



Let's Practice!

Agile Software development is easy, isn't it?

Instructions: Match column A with column B.

COLUMN A

1. This method describes the principles for software development,
2. The products are tested to know where improvement is needed,
3. There aren't many requirements at the beginning of the project,
4. This is what Mr. Hardy called an evolutionary feature,
5. Software managers should consider using this method,

COLUMN B

- a. are there?
- b. didn't he?
- c. isn't it?
- d. shouldn't they?
- e. aren't they?



Extension Activity

How does it work in real life?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about Agile Software development.
- Brainstorm some ideas about it.
- Write a text of 300 words in which you describe a
- real life application of it.
- Share the information with your partners and instructor.

Self-Assessment

Yes No

- I read about Agile software development.
- I think about Agile software development.
- I write about Agile software development.
- I share the information with my partners.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **feedback loop** (noun): a cyclic process of feedback in which the output eventually influences the input of the process.
e.g. They may request some changes afterward, and the team could resolve to apply other modifications thanks to feedback loops.
- **encourage** (verb): to stimulate or inspire to do something.
e.g. I think software managers should consider using this method as it encourages the interaction of the team, the true requirements of the customer and the confidence in the quality of the product.
- **incremental** (adjective): something that increases or grows regularly.
e.g. They may request some changes afterward and the team could resolve to apply other modifications thanks to feedback loops, this is what's called incremental development.
- **at its core** (expression): something that is central or very important for something else.
e.g. This method describes the principles for software development in which flexibility for easily adapting to changes and collaboration between the team members are at its core.
- **promptly** (adverb): quickly, fast.
e.g. The demands of the market change constantly and they require products to be ready promptly.
- **short time frame** (expression): a short period of time.
e.g. That the agile methodology is iterative, which means that the project is divided into short time frames.



Learning Points

Tag Questions With Affirmative and Negative Statements

We use tag questions for confirming a piece of information that was given previously.

When we are almost sure about a statement, but we want to confirm it anyways, we use a tag question at the end of the sentence.

The tag question is composed by a verb and a pronoun. Both the verb and the pronoun that we are going to use depend on the subject, and the verb tenses you used in the previous statement.

Example: *Agile Software development **is** very important nowadays, **isn't** it?*

For the simple tenses, we use the verb **to be** in the tag if the main verb of the sentence is also the verb **to be**; and we use the auxiliary verb 'to do' if we use any other verb.

Example: *You **prefer** the Agile Software Development method, **don't** you?*

Whenever we have an **auxiliary verb** or a **modal verb**, we are going to use these verbs in the tag question instead.

Examples:

*Agile Software Development **has** become very popular, **hasn't** it?*

*Companies **should** consider implementing the Agile Software Development method, **shouldn't** they?*

*Companies **should have** considered implementing the Agile Software Development method, **shouldn't** they?*

Notice, that when we have both an auxiliary and a modal verb, we will use the modal verb in the tag question.

Usually, when the statement is positive, then the tag question is going to be negative; and when the statement is negative, the tag question is going to be positive.

Example: *The waterfall methodology **is not** as efficient as the agile one, **is** it?*

Finally, the pronoun we use is going to depend on the subject of the statement.

Therefore, we can use 'I', 'you', 'we', 'they', or 'it' in the tag question.

Examples

- *This method describes the principles for software development in which flexibility for easily adapting to changes and collaboration between the team members are at its core, doesn't it?*
- *Then, by the end of each iteration, the products are tested to know where improvement is needed, aren't they?*
- *Now, for this methodology, there aren't many requirements at the beginning of the project, are there?*





Learning Points

Understanding the Philosophy Behind Agile Software Development

The Agile Software development is a method for developing software, where the pressure of the market, the requirements of the client, and the tight schedules can be better managed compared to other methodologies. For this learning point, we are going to take a look at some important characteristics of this method.

Adaptive

Agile Software development method is known to be an adaptive type of method. The main concern of the adaptive methods is being able, as its name states, to adapt quickly to changes. As sometimes the needs of a project may alter, adaptability can be crucial for the succeeding in the project. One important characteristic of working under an adaptive method is that the development team does not plan too far into the future, they basically know the tasks that need to be fulfilled in the short term. Therefore, if there are any changes needed, they can easily reorganize the project.

On the other hand, the predictive methods, focus on planning in detail their future tasks. The big problem with this method is that if there are problems in the early stages of the project, and the planning was not as effective. The entire project is affected as adapting to changes becomes harder for this methods.

Iterative

In the waterfall model, there is a separate phase of testing after the build phase has finished. On the other hand, in the Agile development, testing is constantly taking place at the end of every iteration (an iteration is a small task that has to be accomplished in a short period of time) instead of having just one testing phase toward the end of the project. By frequently testing the product, the users are going to get a better understanding of where the product is going and if its features are as expected. This feedback allows the development team to make modifications easily and more accurately adapt the product to the users' needs.

Working software

Different from other methodologies in which the customer is only able to regard the process of the products in a comprehensive document, agile allows the client to test the progress of the product itself. This permits both the customer and the team to get more accurate feedback into how the product is evolving and what possible changes may be necessary.

To conclude, Agile Software development method helps companies to be more flexible toward clients' requests and to adapt to changes easily. Moreover, it also requires frequent communication between the team members and a continuous testing of the product which makes it respond better to what the customer has in mind and be more efficient for the developers.

